RUT/M1a Reviewer Summary: Stage C + Sprint D

Status: Preliminary research summary for review, replication, and falsification-oriented development. Model: Reverse Universe Theory / M1a frozen late-time bridge. Author: Mark "Question Mark" Van Dyk.

1. Purpose and Frozen Parameters

This summary documents the current reviewer-facing status of the RUT/M1a perturbation program after Stage C and Sprint D. The claim is not proof of RUT, but that M1a has moved from background-only compatibility into a perturbation-health and action-to-power testing pipeline.

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Frozen background carried forward
z_t = 0.35    width = 0.03    dG = -0.159
H0 = 73.6    Omega_m = 0.248    r_d = 146.7 Mpc
```

2. Stage C Result: Frozen-Background Perturbation Survival

Stage C tested whether the frozen M1a background could survive growth, lensing consistency, and k-dependent perturbation health checks without retuning the background. The leading corridor was:

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Stage C corridor
mu0 = 0.15    Sigma0 = 0.10    k_c = 0.30 h/Mpc    n = 4.0
```

The primary growth result was weak but stable: M1a was slightly BIC-favored over the matched LCDM comparison, but not strongly enough for decisive evidence. The stronger Stage C result was robustness: leave-one-out and subset diagnostics did not favor LCDM, and the k-dependent scan found a large healthy region with no hard failures across tested parameter combinations.

Conservative Stage C interpretation: The frozen M1a bridge survives first-contact perturbation pressure. It is not proven, but it is not immediately ruled out by the tested growth, lensing, or perturbation-health diagnostics.

3. Sprint D Result: First Action-to-Power Bridge

Sprint D asked whether the Stage C corridor could be connected to internal response structure rather than remaining a purely phenomenological healthy region. The resulting pipeline was:

D_RG -> M_AB^-1 -> mu(k,z), Sigma(k,z) -> P(k,z)

D1/D1b introduced a regularized radiation-gravity diagnostic response. D2/D2b replaced the direct response with an inverse field-response matrix. The strongest D2b result forced explicit nonzero phase-bridge mixing and still reproduced the Stage C corridor:

D2b nonzero mixing	m_phi^2 = 2.0; m_chi^2 = 2.0; lambda_mix = 0.05
Corridor match	mean |Delta mu| ≈ 0.00143; mean |Delta Sigma| ≈ 0.00095; no determinant instability
D3b matter-power signal	max |Delta P/P| full ≈ 0.00960; target zone k = 0.2-0.3 h/Mpc ≈ 0.00693

The D3b result indicates a sub-1% late-time matter-power signature near k ~ 0.2-0.3 h/Mpc. This is not full CAMB or MGCAMB; it is a controlled diagnostic bridge using the frozen background, D2b-derived mu(k,z), and a linear growth equation.

4. Conservative Interpretation and Next Falsifier

Stage C shows that the frozen M1a bridge survives first-pass perturbation checks. Sprint D provides a first mechanism-linked scaffold showing how the Stage C corridor may arise from a regularized radiation-gravity response, a stable inverse matrix response with nonzero phase-bridge mixing, and a controlled CAMB-lite matter-power signal.

Correct narrow claim: RUT/M1a now has a preliminary action-to-power bridge connecting a healthy perturbation corridor to a stable nonzero-mixing response and a small, testable P(k,z)-level signature. It does not replace covariance-aware datasets, full likelihood analysis, or a full Boltzmann-code implementation.

Next falsification step: Map the D2b response into an MGCAMB/EFTCAMB-style mu(k,a), Sigma(k,a) parameterization or compare the D3b CAMB-lite ratio against compressed DESI/Euclid-like constraints. A clean failure requires revision or rejection of the current corridor; clean survival moves RUT/M1a closer to a serious modified-gravity candidate model.

Suggested GitHub commit: Add Stage C and Sprint D reviewer summary